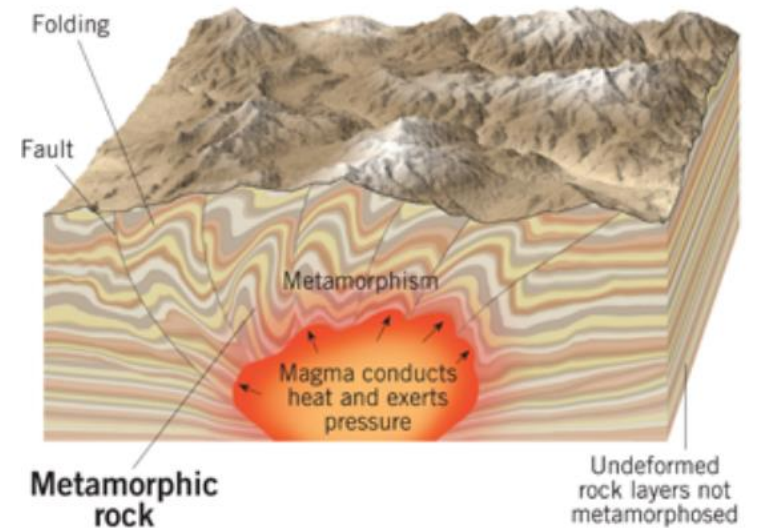
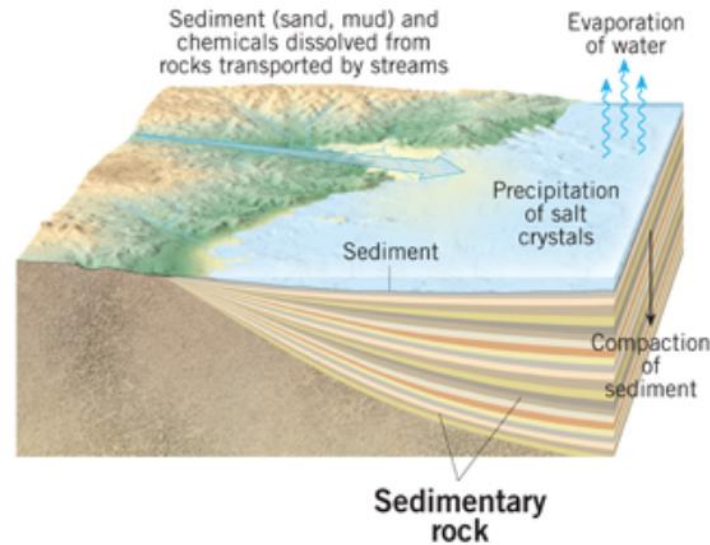
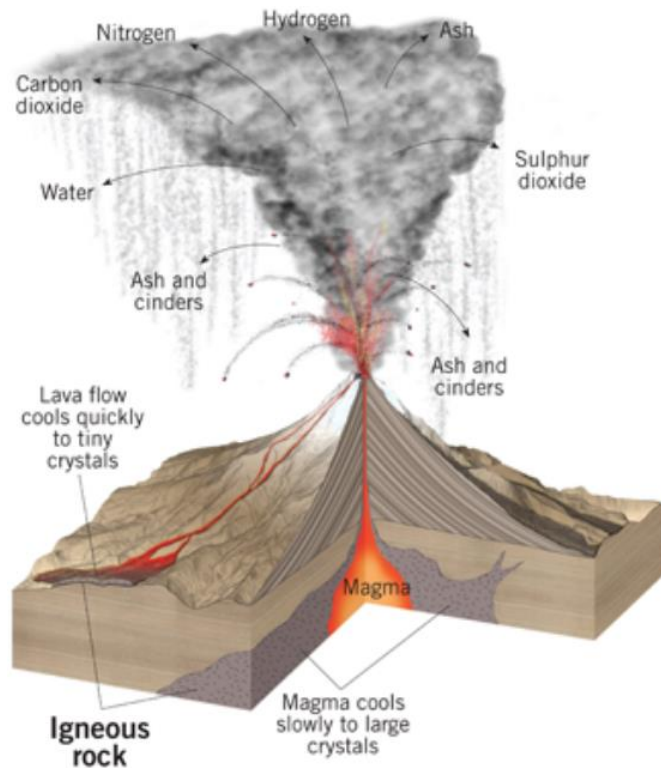


Virtual Rock Kit

Overview

- Geologists divide rocks into three major groups: igneous, sedimentary, and metamorphic



Overview

- Despite the difference in how these three rock types form, they are identified the same way – by characterizing their:
 - Texture – the overall appearance of a rock based on the size, shape, and arrangement of its component(s)
 - Composition – the components (minerals and/or clasts) that make up the rock

Objectives

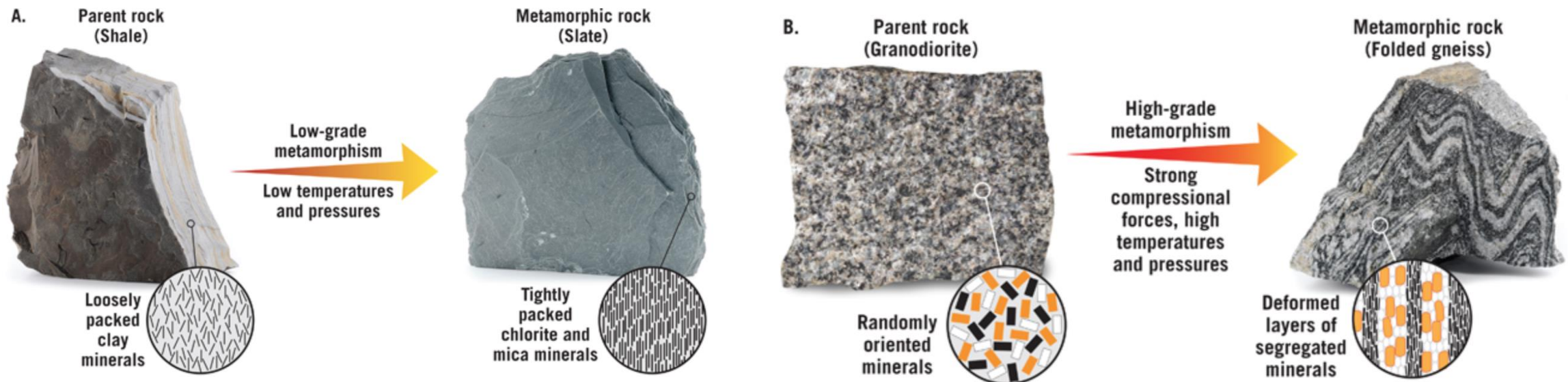
- Characterize the texture and composition of igneous, sedimentary, and metamorphic rocks in your virtual rock kit;
- Identify their proper rock name using the provided identification tables.
 - The rock names highlighted and in bold uppercase font on the identification table are part of this virtual rock kit:

EXAMPLE

Metamorphic Rocks

Metamorphic Rock Identification

- Metamorphic rocks are produced from preexisting sedimentary, igneous, or other metamorphic rocks (these are known as the *parent rock*)
 - The parent rock is subjected to new conditions (higher temperatures and/or pressures) that change the mineralogy (composition) and/or texture of the rock



Metamorphic Rock Identification

- Metamorphic rock's texture tells us the deformational (stress and strain) history of the rock
 - Foliated – Mineral grains align *parallel* to one another and *perpendicular* to the direction of greatest compressional stress

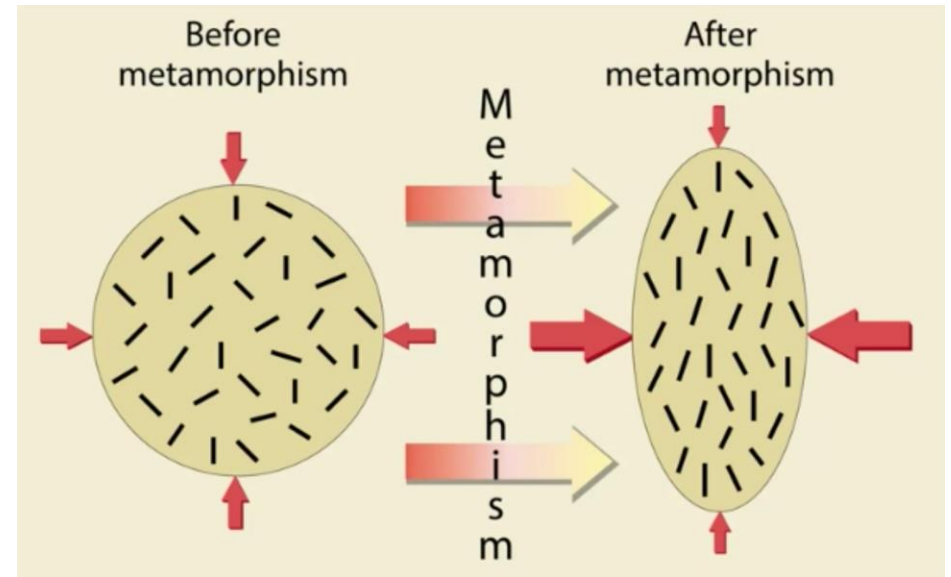


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Metamorphic Rock Identification

- Metamorphic rock's texture tells us the deformational history (or lack thereof) of the rock
 - Nonfoliated – No layering or banding of minerals; mineral crystals interlock

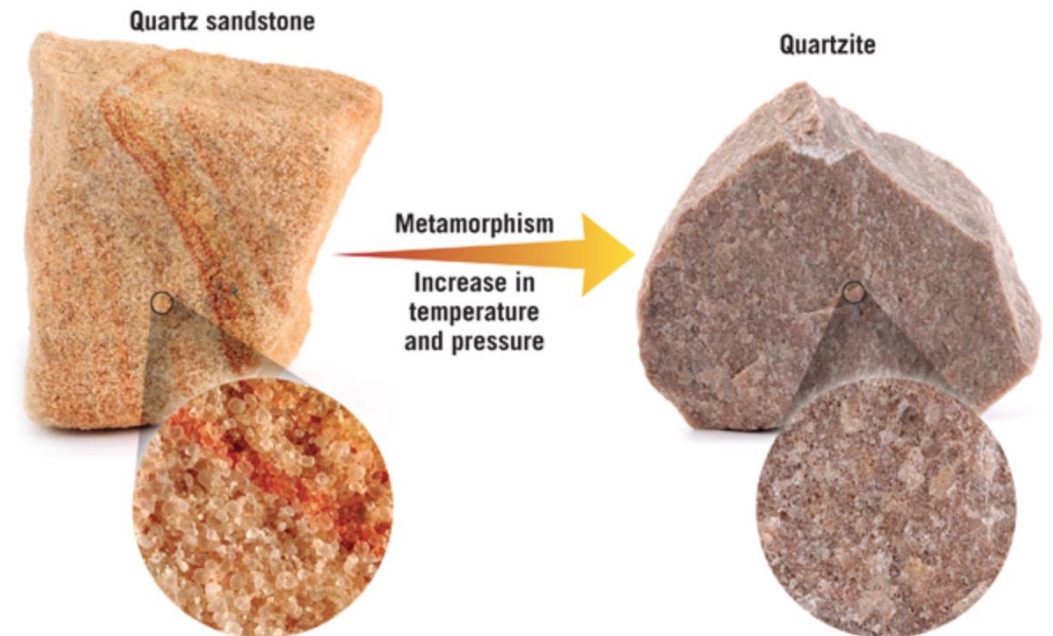


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Metamorphic Rock Identification

- Metamorphic rock's composition is dictated primarily by the *parent rock* composition
 - Quartzite -> Quartz sandstone (both composed of the mineral quartz)
 - Marble -> Limestone (both composed of the mineral calcite)
 - Gneiss -> Granite (both composed of the minerals quartz, feldspars, and biotite)

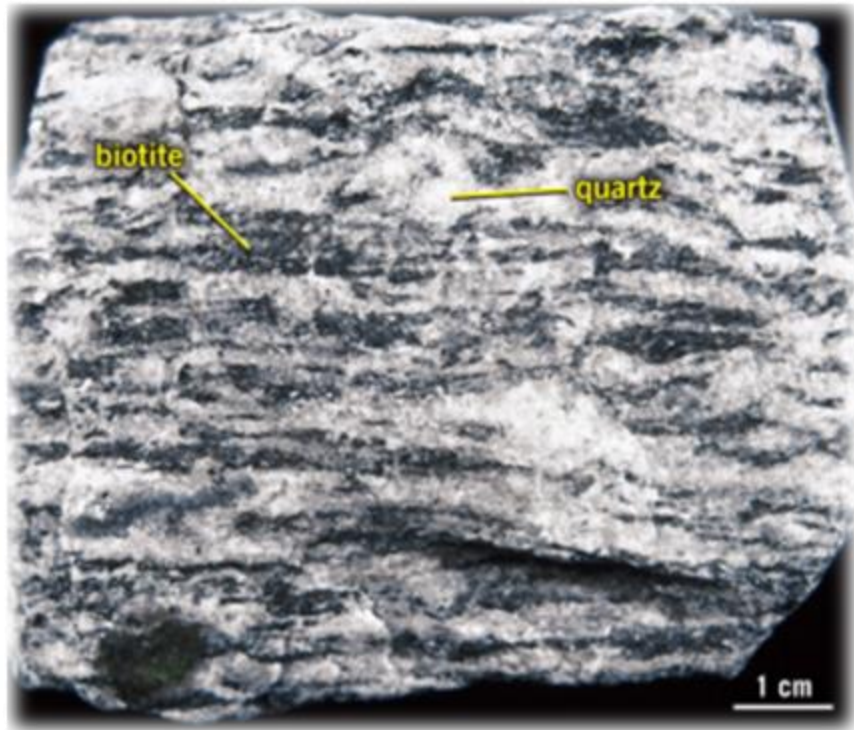
Metamorphic Rock Identification

	Texture	Composition	Rock Name	Parent Rock	Uses
Nonfoliated	Glassy texture	Black glossy rock that breaks along uneven or conchoidal fractures	ANTHRACITE COAL	Peat, Lignite, Bituminous Coal	Highest grade coal (burned for energy)
	Sandy texture	Quartz sand grains fused together (will not rub off like sandstone)	QUARTZITE	Sandstone	Construction/ decorative stone
	Microcrystalline (resembling a sugar cube) or medium to coarse crystalline texture	Calcite crystals of nearly equal size and tightly fused together; fizzes in dilute HCl	MARBLE	Limestone	Art carvings, construction/ decorative stone
Foliated	Fine-grained or no visible grains; flat slaty well-developed cleavage	Dull luster; breaks into hard flat sheets	SLATE	Mudstone or Shale	Roofing, floor tile, tabletops
		Slight shine (like Spandex); breaks along foliation surface	Phyllite	Mudstone, Shale, or Slate	Construction/ decorative stone, sources of gemstones
	Medium to coarse grained; strong mineral alignment	Mostly visible sparkling crystals of micas	SCHIST	Mudstone, Shale, Slate, or Phyllite	
	Medium to coarse grained; minerals alternate (light and dark banding)	Visible crystals of two or more minerals in alternating light and dark foliated layers	GNEISS	Mudstone, Shale, Slate, Phyllite, Schist, Granite, or Diorite	

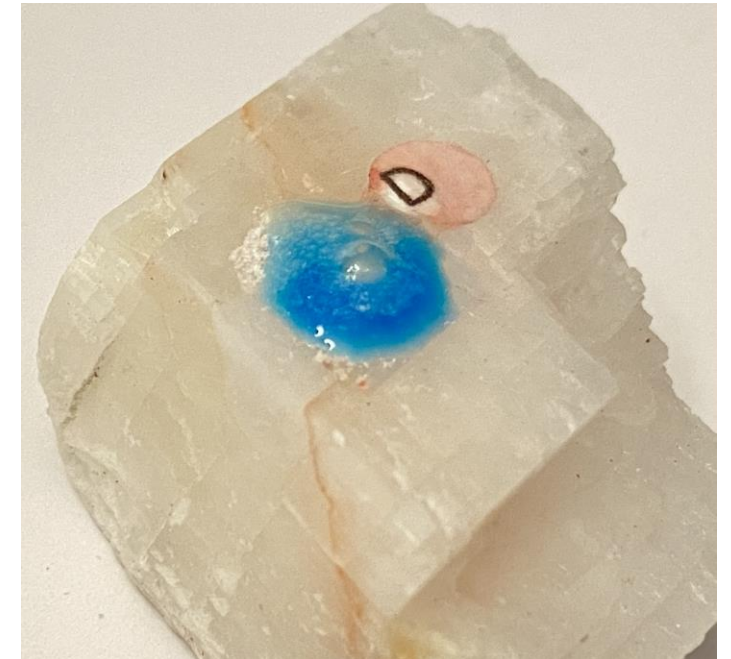
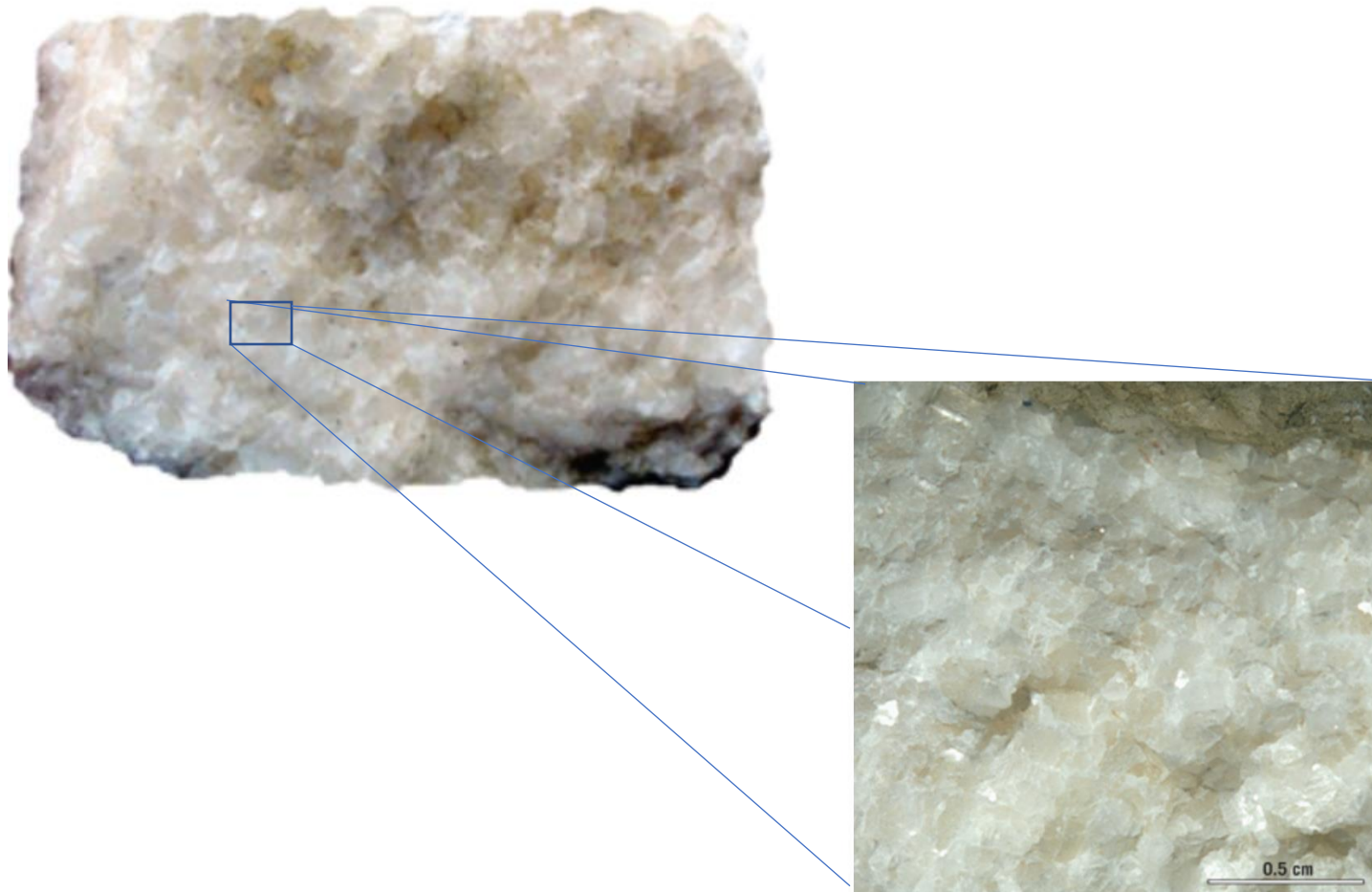
Metamorphic Rock Identification

1. Several examples of the same rock will be shown to you.
2. Observe its metamorphic texture (foliated? or nonfoliated?).
3. Use the identification chart to help determine composition (mineral hints provided when needed).
4. Identify the metamorphic rock name based on the texture and composition observations.
5. Record the "Parent Rock" and "Uses" information in your chart in Lab #2.

Metamorphic Rock Sample “M”

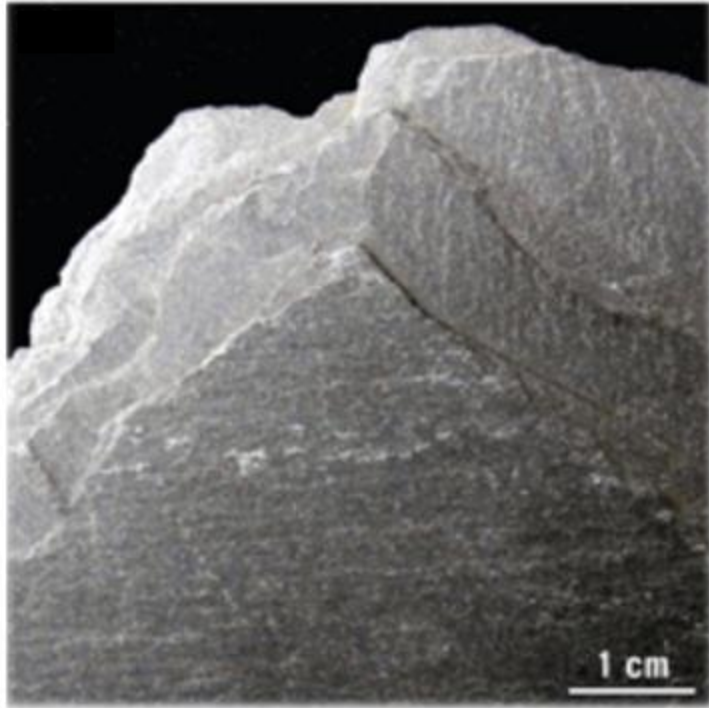


Metamorphic Rock Sample “N”

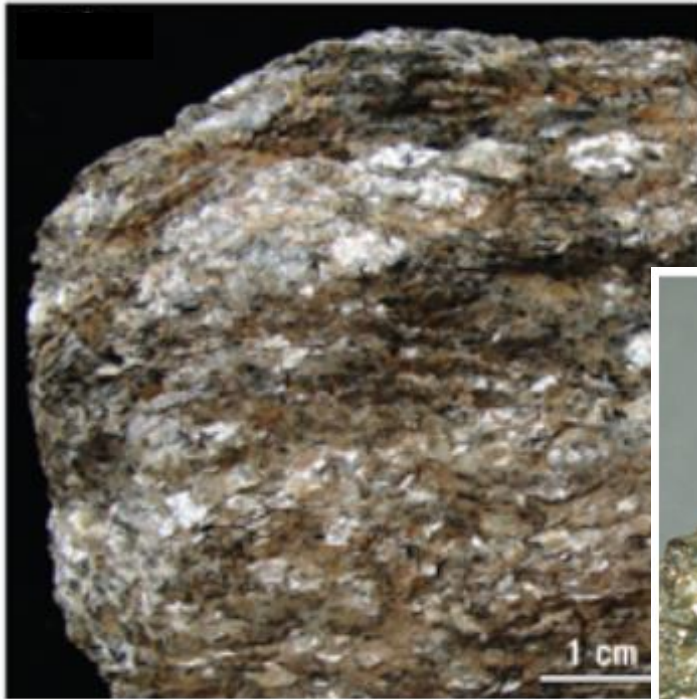


*Mineral information:
Composed of Mineral D from mineral
identification lab*

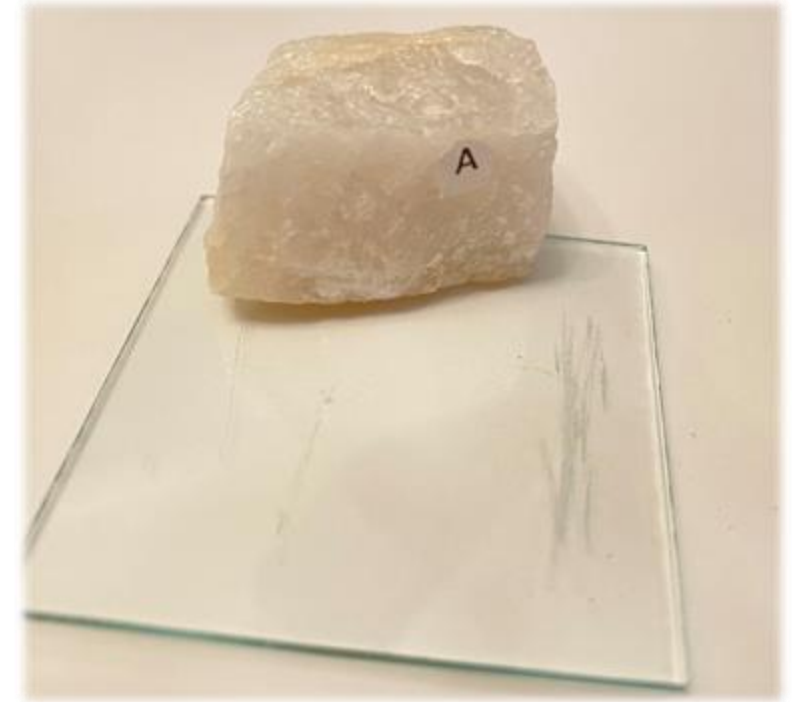
Metamorphic Rock Sample “O”



Metamorphic Rock Sample “P”



Metamorphic Rock Sample “Q”



*Mineral information:
Composed of Mineral A from mineral
identification lab*

Metamorphic Rock Sample “R”

